Christ's College, University of Cambridge

Advert Assignment Auctions

Hugo Kai Jun, SUN

Essay setter: Dr. Frank Kelly

July 6, 2016

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Abstract

We give a brief introduction and comparison of the two mechanisms widely used in sponsored advert auctions, the Vickrey-Clarke-Groves auction (VCG) and the Generalised Second Price auction (GSP). VCG is not manipulable and encourages truthtelling while GSP does not, but both contain a Nash equilibrium where allocative efficiency is achieved and platform revenues are the same. Zero-revenue equilibria are possible in both mechanisms. Conflation removes such zero-revenue equilibria and retains the allocative efficient equilibrium. User heterogeneity results in inefficient allocation for conflated VCG, but not necessarily for conflated GSP. A Bayesian approach where advertisers are aware of the distributions of her opponents' bids gives rise to more complicated behaviour in the GSP. Bid shading may result in a decrease in platform revenue following an increase in click-through rates, or in certain cases renders any efficient equilibria nonexistent.

Introduction

Providing online advertising services have been a major source of revenue for not only search platform companies such as Google, Yahoo! and Microsoft, but also increasingly so for social media platforms like Facebook or LinkedIn. In 2009, Google made 22.89 billion USD from advertising. In 2015 this figure has increased to 67.39 billion USD, which accounts for over 90% of their revenue and 30% of worldwide digital advert revenue [1]. Over the same period of time, Facebook also saw a significant increase of revenue from 0.76 billion USD to 17.08 billion USD [2].

For search platforms, advertisers bid for specific search keywords, and the most sought-after keywords can be worth hundreds of dollars per click[3]! In a more general setting, the query types that advertisers bid on can be users who came from a particular geographic location, or users who have certain groups of acquaintances on Facebook.

The three parties: the users, the platform and the advertisers have conflicting interests. First, users want relevant adverts, not too much to an extent of

[1] `https://abc.xyz/investor/news/earnings/2015/Q4_google_earnings/index.html`
[2] `http://investor.fb.com/releasedetail.cfm?ReleaseID=952040`
[3] `https://searchenginewatch.com/2016/04/14/the-top-100-most-expensive-keywords-in-the-uk/`

becoming a distraction from the main content. Second, advertisers generate revenue by site visits, so they seek to deliver their content to the right clientele with minimal costs. Number of clicks to their adverts is a good proxy of revenue to advertisers, so bids are usually expressed in *dollars per click*. This is different from traditional media such as billboards or newspapers, where bids are in *dollars per impression*. The platform wishes to maximise their revenue from advertisers, and doing so requires a delicate balance between bargaining a bigger share of the advertisers' utility (i.e. to get a bigger piece of the pie), and long term prosperity of the platform by delivering relevant content to the users. Meanwhile the platform has to stay price competitive against other competing platforms, and leave a fair share of utility for the advertisers to produce high-quality adverts appealing to users (i.e. making the pie bigger).

Chapter 1 covers the two mechanisms commonly used by the platforms: *Vickrey-Clarke-Groves auction* (VCG) used by Facebook [4] and *Generalised Second Price auction* (GSP) used by Google [5] and Bing [6]. We shall describe their properties with a brief discussion on their merits and drawbacks. *Chapter 2* is a theoretical description of the effect of *conflation* on VCG and GSP. Conflation refers to treating similar but distinct products as identical, in order to make markets thick [7]. Then the paper seeks to explore different extensions in the analysis of the two mechanisms. In *Chapter 3* we shall look at the *incomplete information setting*, where advertisers are unaware of the value of other competitors. We will look at the motivation for considering this different framework, explore its effect on *bid shading*, and how advertisers will likely react strategically in face of unknown market conditions.

[4] `https://www.facebook.com/business/help/494368557244384/`

[5] `https://support.google.com/adwords/answer/1704424?hl=en-GB`

[6] `http://advertise.bingads.microsoft.com/en-uk/blog/27821/bing-ads-auction-explained-how-bid-cost-per-click-and-quality-score-work-together`

[7] cf. thin market - a thick market has a large number of market participants, price tends to be less volatile with increased liquidity. In general, efficient equilibrium is more reliably achieved and no market participant has monopoly power

Chapter 1

VCG and GSP

We follow a similar notation and framework as in the paper of Kelly et al. [1]

We begin by setting up some notations. A limited set of adverts is shown in response to users submitting a query type. It can be a user searching a particular set of keywords, or someone logging into her social network account. Let $i \in I$ index the finite set of advertisers, $\tau \in T$ index the query types, a potentially infinitely large set. Let $l \in L$ be the set of slots available, which is assumed to be constant and not dependent on i or τ. In the context of a search engine, T contains individual keywords that advertisers can bid for. In a more general context, it can include information of the user such as search history, location, demographic data, group of acquaintances etc., basically any data the platform may get hold of. Regarding how finely the platform shall partition the query type, more discussion follows in *Chapter 2*.

Now let $p_i^\tau \alpha_l$ be the probability of a click-through when advert i is shown in slot l, which we assume to be known by both the platform *and the advertisers*. We can see p_i^τ as the *relevance effect* and α_l as the *slot effect*. For now the two variables are independent. For sake of simplicity, we index the slots in a way such that the click-through rate decreases, so slot 1 is the best slot. Renormalise p_i^τ and α_l such that $\alpha_1 = 1$.

Advertiser i submits a single bid $b_i^* \in \Theta_i$, which reflects his value of a click-through $b_i \in \Theta_i$ but not necessarily the same (although in VCG they are). This value is constant across different types τ or slot position l. θ_i may be taken to be the set of non-negative reals, or with discrete values such as whole number of cents. Let $b = (b_i)_{i \in I} \in \Theta$, the product space of all Θ_i. We assume that no two advertisers have same valuation of a click, which would occur almost surely if they have private values drawn from a distribution that admits a probability

density.

A search query τ triggers an optimisation problem by the platform, who has to determine the allocation and pricing of the adverts. Let $x^\tau_{il}(b^*)$ be the indicator function of advertiser i being assigned on slot l. Let $\pi^\tau_{il}(b^*)$ be the price charged by the platform to advertiser i when his advert on slot l receives a click-through. Obviously π^τ_{il} only makes sense when $x^\tau_{il} = 1$. Let $y^\tau_i(b_i) = y^\tau_i(b_i, b_{-i})$ be the click-through rate advertiser i gets by bidding b_i. With a slight abuse of notation, define $\pi^\tau_i(b_i) = \pi^\tau_i(b_i, b_{-i})$ to be the expected payment per click for advertiser i when she bids b_i. We emphasis the dependence of x^τ and π^τ on b.

We expect advertisers to respond to the function x^τ and π^τ by submitting a bid $b^{\tau *}_i$ that would maximise their private welfare $u^\tau_i((b^*_i, b_{-i}), b_i) = (b_i - \pi^\tau_i(b^*_i, b_{-i}))y^\tau(b^*_i, b_{-i})$. (utility = (value - cost)×click-through rate).

u^τ_i denotes the private welfare for advertiser i. The first argument (b^*_i, b_{-i}) is the bid vector for all advertisers and the second argument b_i is the true value of advertiser i.

1.1 VCG maximises social utility and encourages truthful bidding

VCG aims to maximise social welfare. We have the following problem:

Assignment(τ, b)

$$\text{Maximise} \qquad \sum_{i\in I} b_i \sum_{l\in L} p^\tau_i \alpha_l x^\tau_{il} \tag{1.1}$$

$$\text{subject to} \qquad \sum_{i\in I} x^\tau_{il} \le 1, \quad l \in L \tag{1.2}$$

$$\sum_{l\in L} x^\tau_{il} \le 1, \quad i \in I \tag{1.3}$$

$$\text{over} \qquad x^\tau_{il} \ge 0, \quad i \in I, l \in L \tag{1.4}$$

The two constraints prevent one slot from containing more than one advert, or one advert being placed in multiple slots. This class of LP problem guarantees an integral solution[1] which may be computed efficiently [12, Lecture 6]. One may show that slots are allocated in descending order of the *adjusted bid*

[1] The set of optimal solutions admits one which satisfies $x_{il} \in \{0, 1\}$ for all $i \in I, l \in L$. In other words the allocation of slots need not be randomised except in the case of ties.

$b_i p_i^\tau$. Ties are broken by randomly assigning the slot to the advertisers involved.

Let the click-through rate by player i be:

$$y_i^\tau = \sum_{l \in L} p_i^\tau \alpha_l x_{il}^\tau \tag{1.5}$$

VCG adopts the following payment policy:

Pricing(τ, b)

$$\pi_i^\tau = b_i - \frac{\sum_{j \in I} b_j y_j^\tau(b) - \sum_{j \in I, j \neq i} b_j y_j^\tau(0, b_{-i})}{y_i^\tau(b)} \tag{1.6}$$

Note that the payment by advertiser i is never more than his bid b_i.

Theorem 1.1.1. *VCG admits an equilibrium that is strategyproof and maximises social welfare, i.e. (allocative) efficient.*

Definition. A *(Nash) equilibrium* is a solution of a game where if each player knows the strategy of the other players, he has no incentive to change his own strategy.

Definition. *Strategyproof* means that the mechanism is not manipulable. In this case it means that it is a Nash equilibrium for each agent to truthfully report his true type. That is:

$$u_i((b_i, b_{-i}), b_i) \geq u_i((b_i^*, b_{-i}), b_i), \quad \text{for all} \quad b_i^* \in \Theta_i$$

Proof. The solution where all advertisers bid truthfully and allocating slots in decreasing order of *adjusted bid* $b_i p_i^\tau$ obviously maximises social utility by the Rearrangement Inequality [2]. It remains to show that this solution is stable, i.e. a Nash equilibrium.

$$\begin{aligned}
u_i((b_i^*, b_{-i}), b_i) &= (b_i - \pi_i^\tau(b_i^*, b_{-i})) y_i^\tau(b_i^*, b_{-i}) \\
&= (b_i - (b_i - \frac{\sum_{j \in I} b_j y_j(b_i^*, b_{-i}) - \sum_{j \in I, j \neq i} b_j y_j(0, b_{-i})}{y_i^\tau(b_i^*, b_{-i})})) y_i^\tau(b_i^*, b_{-i}) \\
&= \sum_{j \in I} b_j y_j(b_i^*, b_{-i}) - \sum_{j \in I, j \neq i} b_j y_j(0, b_{-i}) \\
&\leq \sum_{j \in I} b_j y_j(b_i, b_{-i}) - \sum_{j \in I, j \neq i} b_j y_j(0, b_{-i}) \\
&= u_i((b_i, b_{-i}), b_i)
\end{aligned} \tag{1.7}$$

[2] $\sum b_i p_i^\tau \alpha_i \geq \sum b_i p_i^\tau \alpha_{\sigma(i)}$ for any permutation σ if $b_i p_i^\tau$ and α_i are descending

The inequality follows from the fact that $\sum_{j \in I} b_j y_j(b)$, the social welfare, is maximised by truthful reporting. □

Equation 1.6 may look daunting, but in fact it has interesting connection with the concept of *externality* in Economics. In the VCG mechanism, each advertiser basically pays the sum of social welfare he has deprived other players of by his bid. We may modify the VCG mechanism to accommodate advertisers with different utility functions, as long as they satisfy certain monotonicity conditions [1, p.4].

The VCG has a nice equilibrium where social welfare is maximised. However this is not necessarily the only equilibrium of VCG.

Example. *Consider two players bidding for one slot in VCG. Alice bids a very high price (higher than Ben's private value of the slot) and Ben bids zero. This constitutes as a Nash equilibrium, where Alice gets the slot and pays zero. Alice has no incentive to change her bid and the same goes for Ben, for if he tries to outbid Alice he has to pay for the slot a price that is beyond his value.*

The existence of a zero-revenue equilibrium is bad news for the platform. Luckily it does not happen when there is more than one slot. However, if we add a small but non-zero bidding cost $\epsilon > 0$, which may be imposed by the platform but can also be viewed as a naturally occurring transaction cost, then the last slot is always sold at zero price.

Theorem 1.1.2. *For $n > k > 2$, VCG mechanism with k slots and n players with a small bidding cost $\epsilon > 0$ does not have a zero-revenue equilibrium. The last slot is sold at price zero but all other slots have positive prices. This equilibrium persists even when we set $\epsilon = 0$.*

Proof. VCG mechanism determines the pricing from the losing bids. However due to the bidding cost any losing bid would not have bid at the start (i.e. bidding zero). There will only be k bidders with non-zero bids so the last slot is always sold at a zero price. On the other hand, if both the last and second last slots are sold at zero, it means that some player i bids p_i, gets the second last slot and pays zero; some other player j bids 0 and gets the very last slot. Now any other players who have not got a slot has the incentive to outbid player j to secure the last spot paying a price of zero. As a result the second last slot will no longer be free. Contradiction. □

1.2 GSP charges each player the next highest bid

GSP allocates slots to advertisers similarly, by sequentially assigning the best slot to the highest-paying bidder. For all analysis in GSP, we assume bidders are ranked in decreasing order of their bids (*not* necessarily their true valuations) of a click-through. So for all $i \leq k$ the number of slots, advertiser i gets slot i. We use b_i to denote the bid by player i but v_i the true value of a click (not b_i^* and b_i as in VCG, because the two are usually different in GSP). Let α_l denote the click-through rate of the slots.

Player i is charged for each click an amount of:

$$\pi_{il}^{\tau} = \delta_{il} b_{i+1} \quad \text{for } i \leq k \tag{1.8}$$

Theorem 1.2.1. *Truthtelling is not a dominant strategy in GSP. It is also not strategyproof.*

Proof. Suppose there are two slots on a page and three advertisers. An advert in the first slot has click-through rate of 200 and 199 clicks per hour. The three players value a click-through at \$10, \$4 and \$2 respectively. If all players bid truthfully, then player 1 and 2 gets the slot 1 and 2 respectively, paying \$800 and \$398 per hour. Player 1's payoff is equal to $(\$10 - \$4) \times 200 = \$1200$. If, instead, he shades his bid and bids only \$3 per click, he will get the second slot, and his payoff will be equal to $(\$10 - \$2) \times 199 = \$1592 \geq \$1200$.

Hence truthtelling is not a Nash. □

Remark. *If players submit the same bid in the two mechanisms, the GSP always generate higher revenue for the platform.*

Now we demonstrate that GSP contains an efficient equilibrium with the same pricing as in VCG.

> **Strategy profile B**: Player $i \in \{2, ..., \min\{n+1, k\}\}$ submits bid b_i that is equal to the dollar per click amount player $i-1$ would have to pay if they were in the truthtelling equilibrium of the VCG. Player 1 just submits his true value $b_1 = v_1$. Player $i > k$ submits zero.

Example. *As in the proof of Theorem 1.2.1, with two slots and three players. The VCG pricing for player 1 and 2 would be* $(1 \times \$4 + 199 \times \$2)/200 = \$2.01$ *and* $(199 \times \$2)/199 = \$2$ *per click respectively. Therefore player 1 would bid his true value \$10, player 2 would bid VCG price \$2.01 and player 3 would bid*

VCG price \$2, and we end up with the same allocation and payment as in the VCG mechanism.

We can easily check that this configuration is a Nash as no players have the incentive to alter their bid. In fact we can make a stronger statement in the general case:

Theorem 1.2.2. *[4, Edelman et al., p250, Theorem 1] Strategy profile B is a symmetric Nash equilibrium (SNE) of this game. In this equilibrium, each advertiser's position and payment are equal to those in the dominant-strategy equilibrium of the game induced by VCG. In any other SNE of this game, the total revenue of the seller is at least as high as in B.*

Definition. *Strategy dominance* is when one strategy is better (not worse off) than all other strategies for one player, no matter how his opponents act. We have *dominant-strategy equilibrium* when strategy dominance occur for all players.

Each player bidding his own true value is a dominant-strategy equilibrium. Because no matter what the other players bid, a player cannot gain by misrepresenting his bid.

Definition. A *symmetric Nash equilibrium* (SNE) is a Nash equilibrium where intuitively each player "does not want to swap places with the player right above him". It is also known as a *locally envy-free Nash equilibrium.*

$$\alpha_i(v_i - b_{i+1}) \geq \alpha_{i-1}(v_i - b_i), \quad \text{for all} \quad i \leq \min\{n+1, k\} \tag{1.9}$$

Definition. If players can choose their actions randomly, then a *pure strategy* is a strategy that chooses some action with probability 1. Otherwise it is a *mixed strategy.*

Lemma 1.2.3. *The outcome of any NSE of the GSP auction is a stable assignment.*

By *assignment* we mean a matching from the set of available slots L to the set of players I. A stable assignment is when no players can gain by rematching (swapping with another player).

Proof of Lemma 1.2.3. By definition, in any SNE, no advertiser can profitably rematch with the position of the advertiser right above her. Also, no advertiser i can profitably rematch with the position assigned to the advertiser below her $j > i$. For if this rearrangement exists, advertiser i would find it profitable to slightly underbid advertiser j and get his position and payment, contradiction with the assumption of equilibrium.

It remains for us to show that no advertiser can profitably rematch with the position assigned to the advertiser that is more than one spot above her - simply outbidding him slightly will get us his position but not his payment.

First, observe that in NSE the resulting matching must be *efficient*, which means that for all $l \in \{1, ..., k-1\}$ the advertiser assigned to slot l always have a higher valuation than the advertiser assigned to slot $l+1$, hence slots are assigned in descending order in valuation, and we achieve social efficiency. This is proven below:

Let $g : L \to I$ be an assignment function in any SNE. Indeed, if $v_{g(l)}$ and $v_{g(l+1)}$ are the values of advertisers $g(l)$ and $g(l+1)$ assigned to positions l and $l+1$, we have

$$\alpha_l(v_{g(l)} - b_{l+1}) \geq \alpha_{l+1}(v_{g(l)} - b_{l+2}) \tag{1.10}$$

for equilibrium means that player $g(l)$ does not want to move down (which she can by slightly underbidding the person below), and

$$\alpha_{l+1}(v_{g(l+1)} - b_{l+2}) \geq \alpha_l(v_{g(l+1)} - b_{l+1}) \tag{1.11}$$

by SNE.
Summing the two inequalities and cancelling out $\alpha_l b_{l+1}$ and $\alpha_{l+1} b_{l+2}$, we have

$$\begin{aligned} \alpha_l v_{g(l)} + \alpha_{l+1} v_{g(l+1)} &\geq \alpha_l v_{g(l+1)} + \alpha_{l+1} v_{g(l)} \\ (\alpha_l - \alpha_{l+1}) v_{g(l)} &\geq (\alpha_l - \alpha_{l+1}) v_{g(l+1)} \\ v_{g(l)} &\geq v_{g(l+1)} \end{aligned}$$

The last line is because α_l is decreasing.

Now we go back to proving that no advertiser can profitably rematch with an advertiser more than one spot above her. Suppose advertiser $g(l)$ assigned to position l wishes to rematch with $g(m)$ at position $m < l-1$, then by NSE we have:

$$\begin{aligned} \alpha_l(v_{g(l)} - b_{l+1}) &\geq \alpha_{l-1}(v_{g(l)} - b_l) \\ \alpha_{l-1}(v_{g(l-1)} - b_l) &\geq \alpha_{l-2}(v_{g(l-1)} - b_{l-1}) \\ &\vdots \\ \alpha_{m+1}(v_{g(m+1)} - b_{m+2}) &\geq \alpha_m(v_{g(m+1)} - b_{m+1}) \end{aligned}$$

Now for each inequality, we may replace $v_{g(p)}$ with $v_{g(l)}$ for each $p \in \{m+1, ..., l\}$, the inequality remains valid as we have:

$$\alpha_p(v_{g(l)} - b_{p+1}) - \alpha_{p-1}(v_{g(l)} - b_p) = (\alpha_p - \alpha_{p-1})(v_{g(l)} - v_{g(p)}) + (\alpha_p(v_{g(p)} - b_p) - \alpha_{p-1}(v_{g(p)} - b_p)) \geq 0 \tag{1.12}$$

The first term is non-negative as α_l is decreasing and $v_{g(i)} \leq v_{g(l)}$, and the second term is also non-negative from the inequalities shown above. Now we sum up all the inequalities and we get:

$$\alpha_l(v_{g(l)} - b_{l+1}) \geq \alpha_m(v_{g(l)} - b_{m+1}) \tag{1.13}$$

Therefore it is never profitable to rematch to position m. □

It is reasonable to restrict ourselves to the set of Nash equilibria which are also SNE. If this condition is violated, i.e. for example player i prefer the slot $i-1$ over her own, she can raise her bid slightly, not affecting her own payoff but punishing the player on the next slot, say player j. Player j can retaliate, and the most he can do is to slightly underbid player i, effectively swapping place with her. [3]

Proof of Theorem 1.2.2. Let b_i^* denote the bid of player i when he adopts strategy profile B. Let $P^{V,(i)}$ denote the payment (click rate multiplied by click-through probability) in the VCG equilibrium by player i, so for $j \geq 2$ player j submits a bid of $\frac{P^{V,(j-1)}}{\alpha_{j-1}}$. Note that the difference between $P^{V,(j-1)}$ and $P^{V,(j)}$ is precisely the externality imposed on player j by player $j-1$, that is $(\alpha_{j-1}-\alpha_j)v_j$.

First we check that the order of the bids is preserved and we achieve allocative efficiency, i.e. $b_j^* > b_{j+1}^*$ for any $j < \min\{n, k\}$. That is for $j \geq 2$:

$$\begin{aligned}
& \frac{P^{V,(j-1)}}{\alpha_{j-1}} > \frac{P^{V,j}}{\alpha_j} \\
\iff \quad & \frac{(\alpha_{j-1} - \alpha_j)v_j + P^{V,(j)}}{\alpha_{j-1}} > \frac{P^{V,(j)}}{\alpha_j} \\
\iff \quad & \alpha_j(\alpha_{j-1} - \alpha_j)v_j > (\alpha_{j-1} - \alpha_j)P^{V,(j)} \\
\iff \quad & \alpha_j v_j > P^{V,(j)}
\end{aligned}$$

And for $j = 1$,

$$\alpha_1 v_1 > P^{V,(1)}$$

Obviously the VCG payment by player j shall not exceed the value he gains from the clicks $\alpha_j v_j$, or else he could have secured zero payoff by bidding zero.

Next we prove that it is a Nash equilibrium: that no bidder j shall bid a value b' deviating from b_j^*.

[3] Some authors disagree with the notion of disregarding non-SNEs. They argue that there is no clear reason as to why players would retaliate, a player may have good reasons not to increase the price paid by the player on the slot above, as a desire to keep market price low in the long run through tacit collusion. [6, Dütting, p9]

Case 1. $b' > b_j^*$
The advertiser only benefits if this puts him in position $j' < j$. The net payoff from this deviation is given by:

$$\begin{aligned}
&(\alpha_{j'} - \alpha_j)v_j - (\alpha_{j'} b^*_{j'} - \alpha_j b^*_{j+1}) \\
\leq &(\alpha_{j'} - \alpha_j)v_j - (\alpha_{j'} b^*_{j'+1} - \alpha_j b^*_{j+1}) \\
= &(\alpha_{j'} - \alpha_j)v_j - (P^{V,(j'+1)} - P^{V,(j+1)}) \\
= &\sum_{j'}^{i=j-1} (\alpha_i - \alpha_{i+1})v_j - \sum_{j'}^{i=j-1} (\alpha_i - \alpha_{i+1})v_{i+1} \\
\leq &0
\end{aligned} \tag{1.14}$$

The first inequality is because b_j^* is decreasing. The last equality holds since v_i is decreasing in i. Hence deviation is not profitable.

Case 2. $b' < b_j^*$
By under representing her bid, an advertiser is put instead in position $j' > j$. Her payment will be equal to the amount that he would need to pay to be in position j' under VCG. But from our result proven for VCG, truthful bidding is a Nash and a player wold not want to go from his own VCG pricing to someone else's VCG price. Therefore such deviation cannot be profitable.

Now we prove that the equilibrium is indeed SNE. If player j rematches with player $j-1$, his payoff would change by:

$$\begin{aligned}
&(\alpha_{j-1} - \alpha_j)v_j - (P^{V,(j-1)} - P^{V,(j)}) \\
= &(\alpha_{j-1} - \alpha_j)v_j - (\alpha_{j-1} - \alpha_j)v_j \\
= &0
\end{aligned} \tag{1.15}$$

So in fact each player is indifferent between swapping bids with the bidder above, hence the equilibrium is an SNE.

Finally we have to show that B is the SNE that generates the least amount of revenue. Note that any SNE maps us to a stable assignment as proven in Lemma 1.2.3. Suppose this assignment admits a vector of payments $P = (P_1, ..., P_k)$. As before let P^V be the vector of dominant-strategy VCG payments.

In any stable assignment, $P_k \geq \alpha_k v_{k+1}$, otherwise advertiser $k+1$ (who is just shy of getting the lowest slot) would rematch to slot k. $P_k^V = \alpha_k v_{k+1}$, so $P_k \geq P_k^V$.

We proceed inductively. Suppose $P_i \geq P_i^V$ for all $i \geq l$. Then we must have $P_{l-1} - P_l \geq (\alpha_{l-1} - \alpha_l)v_l$ for other wise advertise l would prefer to rematch

with position $l-1$. Therefore

$$\begin{aligned} P_{l-1} &\geq (\alpha_{l-1} - \alpha_l)v_l + P_l \\ &\geq (\alpha_{l-1} - \alpha_l)v_l + P_l^V \quad \text{(induction hypothesis)} \\ &= P_{l-1}^V \end{aligned}$$

Therefore in any SNE the total revenue of the platform is at least $\sum_{j=1}^{k} P_j^V$. □

The zero-revenue equilibria depicted in our first *Example* also applies to GSE, although it is obviously not an SNE. [4]

At this point, we see that VCG and GSP are in fact not too different. First, in both cases the payment of a bidder does not directly depend on his bid, but the bids below him. Nonetheless, the properties of the two mechanisms are rather different. GSP does not have a dominant strategy equilibrium in general, but just a SNE which is guaranteed to generate at least as much revenue as the VCG.

One might therefore wonder as to why VCG is not more widely adapted since it achieves the VCG equilibrium more reliably. Moreover, the mathematical simplicity of VCG makes it a lot more versatile upon generalisation of the problem such as assigning different utility functions to different players and varying the number of slots [1].

[4] In Exapmle 1, one may argue that we still have allocative efficiency (and hence social welfare maximisation) because if Ben has a higher private value than Alice, he has the incentive to bid his true value instead of bidding zero. Then Alice is forced to underbid Ben to cut her loss. Some literature prefers studying *allocative efficient equilibrium* rather than SNE, which forms a strict superset of the set of SNEs [6]

Chapter 2

Conflation

Conflation refers to the act of treating similar but distinct products as identical and is common in the world of commodities trade. In the market for wheat, a typical commodity, standards were created to define wheat grades, where small differences between units are disregarded. For example, in New York, red winter wheat No.2 was required to be "sound, plump, dry, and reasonably clean, contain not more than 10 percent of white winter wheat, and weigh no less than 58 pounds [per bushel]. Winchester standard" (Dondlinger, 224). [5]

Similarly in advert auction, advertisers are allowed to submit a single bid across all available slots. This is an example of conflation as advertisers cannot express their preferences between individual slots, therefore in this chapter we shall refer to the VCG and GSP mechanisms discussed in the previous chapter as *conflated VCG* and *conflated GSP*.

What happens if we introduce some user heterogeneity into the model? Following the route of Milgrom et al. [2], a simple case would be to assume there are two types of users, *buyers* and the *merely curious* with different click-through probabilities. A particular user is a *buyer* with probability $\lambda \in (0, 1)$, and *merely curious* with probability $1-\lambda$. Only clicks from buyers are valuable to the advertisers, but advertisers are unable to differentiate clicks from one type to another.

This would have a distorting effect on the auction. For example, if buyers are more willing to scroll down the page and click on the adverts, then we would expect the lower slots to be more valuable than the upper ones, thus providing advertisers an incentive to shade their bid in order to reduce cost. Therefore we suppose the platform observes one set of click-through probability α_l and the advertisers observe in addition the *relevant* click-through probabilities β_l where $\beta_l \leq \alpha_l$. We would typically expect α_l to decay quicker than β_l as l increases.

This is an example of information asymmetry where advertisers have knowledge of users behaviour after they leave the search platform, information to which the platform does not have direct access.

Suppose the platform knows of the real click-through rates β_l, then they would be able to apply the VCG, allocate slots in descending order of b_i, and charge advertiser i an amount of:

$$P_i^V = \sum_{j=i+1}^{l} b_j(\beta_j - \beta_{j+1}) \tag{2.1}$$

where $\beta_{k+1} = 0$.
First we explain the need for conflation. Consider the unconflated setting of the VCG and GSP, where advertisers submit a bid for *each* slot. We also assume a small bidding cost ϵ naturally imposed on every non-zero bid.

For VCG the allocation is a matching problem to maximise social welfare, where we charge each advertiser according to the externality he has imposed on other advertisers.

For GSP we refer the unconflated setting to *a sequence of second-price auctions*, assignment of slots are decided in the same way as in VCG. Let vector $b_i = (b_{i1}, ..., b_{in})$ denote the bid advertiser i is prepared to make for each of the n positions. WLOG rearrange the bidders so that advertiser i wins the ith slot, then advertiser i pays the second-highest bid among the bidders who have not yet got a slot: payment is $\pi_i = \max_{j>i} b_{ji}$. If bidder i makes J_i positive bids, his payoff is $\alpha_i(v_{ii} - \pi_i) - \epsilon J_i$.

Theorem 2.0.4. *In any pure-strategy equilibrium of VCG auction with bidding cost ($\epsilon > 0$), the platform's revenue is zero. In this equilibrium, for all position l, winner i of position l bids $b_{il} = v_{il}$ for that position and zero for all other positions. This zero-revenue equilibrium persists when bidding cost goes to zero. The same goes for GSP.*

Proof. With the inclusion of a non-zero bidding cost, an advertiser's best response never includes any positive losing bids. It is clear that the strategy profile where each advertiser bids truthfully for the item that would be assigned to her in an efficient assignment, and zero for other items; would constitute as a Nash equilibrium. □

Conflated VCG and GSP both insist that each advertiser submits one bid b_i for all the slots, and we assume that $b_{il} = b_i\alpha_l$ where α is the click-through probability observed by the platform.

Theorem 2.0.5. *Conflated VCG does not result in unconflated VCG pricing (P^V) in the model with two types of searchers.*

Proof. We demonstrate this by means of a counterexample. Suppose there are three slots with overall click-through probabilities $\alpha_1 = 1, \alpha_2 = 0.5, \alpha_3 = 0.4$, but advertisers are aware that the real click-through probabilities are $\beta_1 = 0.5, \beta_2 = 0.45, \beta_3 = 0.4$. There are three advertisers with private values of $v_1 = \$30, v_2 = \$20, v_3 = \$10$.

The unconflated VCG outcome should assign slot 1 to agent 1 at price: $0.05 \times \$20 + 0.05 \times \$10 = \$1.5$, slot 2 to agent 2 at price: $0.05 \times \$10 = \$0.5$, and slot 3 to agent 3 at price zero. Now suppose the same outcome is achieved in conflated VCG by bids $b_1 \geq b_2 \geq b_3$ (so that allocation is efficient). Agent 1 needs to pay: $0.5b_2 + 0.1b_3$ and agent 2 needs to pay $0.1b_3$. For the pricing to match the unconflated VCG, we have $b_2 = \$2$ and $b_3 = \$5$. Contradiction.

□

Here we see that in conflated VCG, bidders make one-dimensional reports, and the platform has too little information to implement full VCG accurately. On the other hand, a GSP auction using conflated message may, in equilibrium, lead to VCG prices.

Theorem 2.0.6. *[2, Milgrom et al., p69, Theorem 7] For the model with two types of searchers, the following two conditions are equivalent:*

1. *There is a pure-strategy Nash equilibrium of the GSP which the assignment is efficient and prices paid by winning bidders are the unconflated VCG prices P^V.*
2. *The VCG price-per-click sequence $\{P_l^V/\alpha_l\}$ is decreasing.*

Proof. Let k be the number of slots. An efficient allocation requires slot l to be assigned to advertiser l. To match pricing with VCG, we need advertiser i to pay a total amount of $P_i^V = \sum_{j=i}^{k} v_{j+1}(\beta_j - \beta_{j+1})$ per click. In order to achieve this, advertiser $i \in \{2, ..., k+1\}$ bids a price of P_{i-1}^V/α_{i-1} (note the same form as stated in the theorem); advertiser 1 bids $v_1\beta_1/\alpha_1$ and advertiser i bids 0 for all $i > k+1$.

$\Rightarrow$: If $\{P_k^V/\alpha_k\}$ is not decreasing, then the bids are not ranked in the right order required for an efficient assignment. More specifically, if $P_j^V/\alpha_j < P_{j+1}^V/\alpha_{j+1}$, then advertiser $j+1$ places a higher bid than advertiser j, hence getting a better slot, despite having lower value, this assignment is inefficient.

$\Leftarrow$: Now suppose $\{P_k^V/\alpha_k\}$ is decreasing, and we wish to show that an advertiser j does not benefit from bidding a value b' deviating from $b_j^* = P_{j-1}^V/\alpha_{j-1}$.

The proof proceeds very similarly to Theorem 1.2.2 so we will just give a brief explanation.

If bidding $b' > b_j^*$ results in the advertiser being placed at a better place $j' < j$, the net payoff is $(\beta_{j'} - \beta_j)v_j - (\alpha_{j'} b^*_{j'+1} - \alpha_j b^*_{j+1}) = (\beta_{j'} - \beta_j)v_j - (P^V_{j'} - P^V_j) = (\beta_{j'} - \beta_j)v_j - \sum_{i=j'}^{j-1} v_{i+1}(\beta_i - \beta_{i+1}) \leq (\beta_{j'} - \beta_j)v_j - \sum_{i=j'}^{j-1} v_j(\beta_i - \beta_{i+1}) = 0$. The last inequality is given by the fact that v_j is decreasing.
If $b' < b_j^*$ resulting in a lower place $j' > j$, $(\beta_{j'} - \beta_j)v_j + \sum_{i=j}^{j'-1} v_{i+1}(\beta_i - \beta_{i+1}) \leq (\beta_{j'} - \beta_j)v_j + \sum_{i=j}^{j'-1} v_j(\beta_i - \beta_{i+1}) = 0$. So again unprofitable. □

Here we see that GSP has the advantage of achieving the unconflated VCG pricing in this situation. But can we show that we do not get any other nasty additional equilibria alongside? To show this, we need to build up some machineries in conflation theorem.

2.1 The Simplification Theorem

We adopt the notation used by Milgrom et al. [2].

Let (I, X, ω) be a mechanism where I is the set of players, $X = (X_1, ..., X_I)$ is the set of strategy profiles, $\Omega \in \Omega_1 \times ... \times \Omega_I$ is the set of possible outcomes endowed with a topology. Let $\omega : X \to \Omega$ be the outcome function, and in particular let ω_i be the outcome function concerned by the player i. Each player has their utility function $u_i : \Omega_i \to \mathbb{R}$.

In our case I is the set of advertisers, Ω is the set of feasible allocations of slots and pricing endowed with the *discrete topology*, ω is the mapping from the bids to the allocations, either by VCG or GSP. ω_i is player i's allocation and pricing, and finally u_i is click-through rate × (value - price per click).

Definition. If $\hat{X} \in X$ and $\hat{\omega} = \omega|_{\hat{X}}$, then the mechanism $(I, \hat{X}, \hat{\omega})$ is a *simplification* of (I, X, ω).

Definition. A simplification $(I, \hat{X}, \hat{\omega})$ of (I, X, ω) has the *outcome closure property* if for every player i, every profile $\hat{x}_{-i} \in \hat{X}_{-i}$, every profile $x_i \in X_i$, and every open neighbourhood O of $\omega_i(x_i, \hat{x}_{-i})$, there exists $\hat{x}_i \in \hat{X}_i$ such that $\omega_i(\hat{x}_i, \hat{x}_{-i}) \in O$.

Intuitively it means that the simplification allows a player to achieve the same private outcome as before regardless of the other players' actions. [1]

[1] The inclusion of a topology for ω and open neighbourhood in the definition is for the purpose of completeness, and is not required for the proof of *Theorem 2.1.2*, hence our choice

Definition. For $\epsilon \geq 0$, a strategy profile $x \in X$ is an *ϵ-Nash equilibrium* if for each player i and every strategy $x_i' \in X_i$, we have $u_i(\omega_i(x_i', x_{-i})) - u_i(\omega_i(x)) \leq \epsilon$. A Nash equilibrium is an ϵ-Nash wish $\epsilon = 0$.

Definition. A simplification is *tight* if, for every continuous utility function vector $u = (u_1, ..., u_I)$ and every $\epsilon \geq 0$, every pure strategy profile x that is an ϵ-Nash equilibrium of $(I, \hat{X}, \hat{\omega})$ is also an ϵ-Nash equilibrium of (I, X, ω)

Intuitively, it is saying that our simplification does not introduce any additional ϵ-Nash equilibria.

Theorem 2.1.1. *[2, Milgrom et. al, p66, Theorem 1] Any simplification $(I, \hat{X}, \omega|_{\hat{X}})$ of (I, X, ω) with the outcome closure property is tight.*

Proof. Fix $u \in U$ and $\epsilon \geq 0$, and let simplification $(I, \hat{X}, \omega|_{\hat{X}})$ of (I, X, ω) has the outcome closure property. Now suppose profile $\hat{x}$ is not an ϵ-Nash equilibrium of the (I, X, ω). Then for some player I and strategy $x_i' \in X_I$, $\delta := u_i(\omega_I(x_n', \hat{x}_{-n})) - u_i(\omega_i(\hat{x})) - \epsilon > 0$.

Continuity of u_n and the outcome closure property imply the existence of some $x_n'' \in \hat{X}_n$ such that $u_i(\omega_n(x_i'', \hat{x}_{-i}))$ is in the neighbourhood of $u_i(\omega_I(x_n', \hat{x}_{-n}))$. In particular we can have $u_i(\omega_n(x_i'', \hat{x}_{-i})) - u_i(\omega_I(x_n', \hat{x}_{-n})) < \delta$. So by triangular inequality, $u_i(\omega_i(x_i'', \hat{x}_{-i})) - u_i(\omega_i(\hat{X})) - \epsilon > 0$, so $\hat{x}$ is not an ϵ-Nash equilibrium of $(I, \hat{X}, \omega|_{\hat{X}})$. □

Theorem 2.1.2. *GSP is a tight simplification relative to the (unconflated) VCG mechanism.*

Proof. X is the space of unconflated bidding where players submit a bid for each slot $X = \{b_{il} : i \in I, l \in L, b_{il} \geq 0\}$; $\hat{X}$ only allows one bid from each player and is a subspace of X: $\hat{X} = \{b_{il} = b_i \alpha_l : i \in I, l \in L, b_i \geq 0\}$ for some given α.

By the Simplification Theorem, it suffices to show that for every $i \in I$, every $\hat{x}_{-i} \in \hat{X}_{-i}$, every $x_i \in X_i$, there exists $\hat{x}_i \in \hat{X}_i$ such that $\omega_i(\hat{x}_i, \hat{x}_{-i})$ is arbitrarily close (in fact we will show equality) to $\omega_i(x_i, \hat{x}_{-i})$.

Denote the VCG outcome $\{\omega(x_i, \hat{x}_{-i})\}$ by O, WLOG rearrange players so that player i get slot i where $i \leq n$ in O, so we have one dimensional bids $b_1 > ... > b_{i-1} > b_{i+1} > ...b_n$, and b_i which expresses player i's preference over individual slots. In the simplified setting, player i can secure the same place by bidding an

of the discrete topology.

amount between b_{i-1} and b_{i+1}. Pricing only depends on $\hat{x}_{-i}$ and is therefore the same. □

Here we can see aspects where GSP is advantageous over VCG, as it has the nice property of achieving the dominant-strategy equilibrium of unconflated VCG, while avoiding the zero-revenue equilibria. GSP even achieves this without introducing additional equilibria.

2.2 Discussion

This chapter touches on a deeper topic of information asymmetry between advertisers and the platform. Nowadays, search platforms or social media hold a vast amount of user data from user history across different websites and other platforms, and they have the ability to partition users and queries into much finer type than before. This potentially allows advertisers to deliver much more specific and personalised adverts and create significant value. Information asymmetry between the two parties is in fact two-sided, advertisers have information of what users do after leaving the search platform into their websites, such as whether or not a customer actually makes a purchase. By comparing this data with advert performance stream from the search platform, advertisers can identify indicators of a valuable click such as which keywords and which slot the user clicked on. This data is potentially useful for the search platform to optimise their allocation algorithm so users see more relevant data.

Despite these advantages, this information on keywords and slots is often not passed on to the advertisers by means of conflation. Feedback of advert reception from platform is usually delayed and in the form of aggregated data. In most platforms advertisers are not informed of the winning bid - advertisers only way for them to find out is to win the slot themselves (more to be said in *Chapter 3*). Platforms might indeed have good reasons to do so.

As we saw earlier, absence of conflation jeopardises platform revenue. It leads to thin market where market power skews toward a small number of advertisers, resulting in inaccurate (under) pricing and promote zero-revenue equilibria.

From the perspective of advertisers, overly fine targeting incentivises advertisers to participate in a socially inefficient arm race of cherry-picking the best types. In one extreme example, there was a proposed contract between McDonald's and Yahoo! under which Yahoo! would have shown Happy Meal ads when the sun was shining or the stock market was up. Presumably this "happy contract" would have left rainy days when the market was down for Burger King's or

Wendy's ads. [5]

As a result, advertisers do not have confidence in believing their adverts are delivered to a representative cross-section of the users, instead they may well be left with picked-over impressions. This creates a high barrier to entry and transaction costs in the market as advertisers are constantly monitoring the reception and performance of their adverts. This leads to adverse selection, discourages bidder participation and raises costs for those who do participate.

To some extent, this problem can be alleviated by means of a secondary auction. Query types may be categorised into groups (e.g. it can be "first page of Google search"/"not on first page"; or "Asia"/"Europe" for geographic data of user type), where an auction is conducted to establish a uniform price for each category. It is then partitioned into smaller types where winners of the primary auction bids again for minor adjustment within the category. This combination of practices reduces measurement cost of advertisers (to figure out which type best suits them), encourages thick market and protects advertisers from adverse selection; and has been widely used in real world auction of wholesale diamonds and sales of radio spectrums in government run auctions[5].

Chapter 3

Incomplete Information Setting

This chapter focuses on exploring the more recent advancement and development in sponsored advert auction. We aim to provide a balance between mathematical rigour and conveying the motivation behind different approaches.

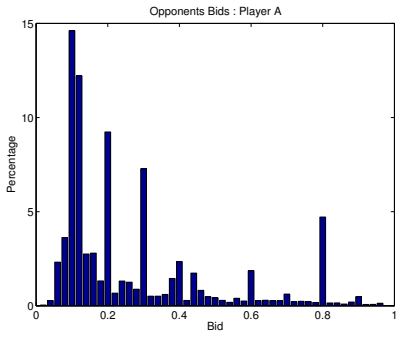
3.1 Motivation

Our analysis up to now focuses solely on ex-post Nash equilibria, where players are aware of each other's type and will never "regret what they have chosen" after the outcome. In reality, each advertiser only has a belief in the prior distributions of other players, which she adjusts as the game is played out repeatedly. One would hope that the outcome of the game shall converge to one of the complete information Nash equilibria, and in particular an SNE.

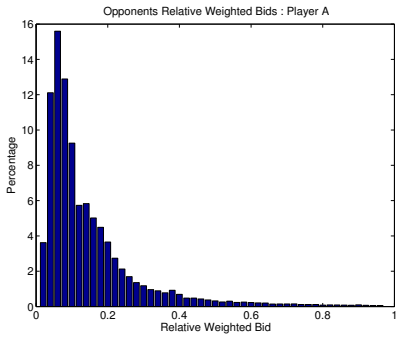
One may question the relevance of incomplete information setting. Advertisers can quickly change their bids in response to market reception. This should result in rapid convergence to the ex-post Nash as market players rapidly adjust their bids. This argument is delivered with some mathematical ground in the paper of Kelly et al. [1, p10] on convergence dynamics.

On the other hand, it has also been argued that advertisers may bid stochastically, and market price at difference time slots can be modelled as i.i.d. drawn from a fixed and unknown probability distribution [8]. This assumption is analogous to classical models in finance, where despite strategic behaviour of traders at the individual level, models of macroscopic price evolution that are stochastic

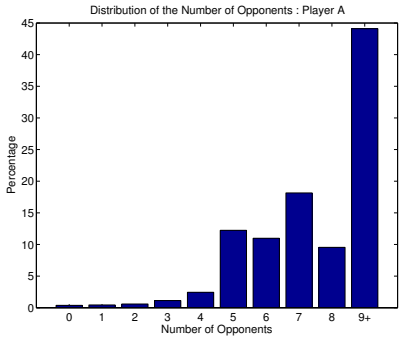
can be effective in developing the algorithms. [10]



(a) Player A's opponents bids



(c) Player A's opponents relative weighted bids



(e) Player A's number of opponents

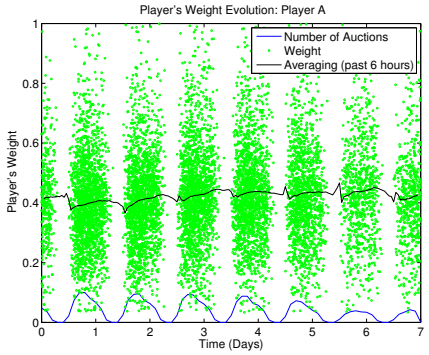
Figure 3.1: Player cannot learn her opponents' bids as their identities change from one auction to another, but rather she learns their distribution. [8, p7, Figure 3].

Observation of real life data shows that a typical advertiser participates with the same bid in a great number of auctions with varying conditions. A player's bid is multiplied by their relevance weighting, which depends on many criteria outside of her knowledge and control. The number and the identities of opponents faced by a given player usually vary across successive auctions. For example, if a player places a bid on the keyword "insurance", she may enter an auction for "house insurance" and another for "car insurance", facing distinct opponents while submitting the same bid. The profile of the user performing the query can change the relevance of the ads. Each successive auction has different opponents, weights and reserve prices.

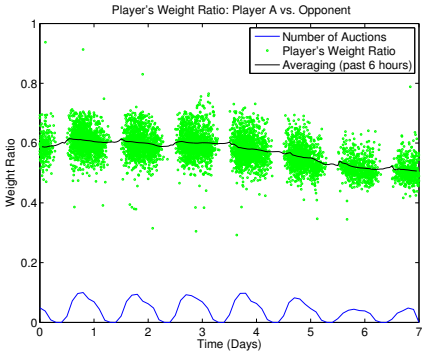
The assumptions we made where a single auction is carried out with full information, or identical auctions played out infinite number of times is questionable. A player cannot learn her opponents' bid or valuations since they change from one auction to another, but rather the distribution of number of opponents and distribution of their bids.

Pin & Key [8] analysed in full record one week of auction ran on Bing of a user on a single keyword. First observation is the variation in the number and identities of opponents faced by one particular player over successive auctions (Figure 3.1).

Figure 3.2: weight-adjusted bid shows variance (a) with standard deviation around 40% of the mean weighted bid. This effect is not entirely explained by the variation in the profile of user performing the query as the ratio of weighted bids still shows standard deviation around 10% of the mean [8, p5, Figure 1].



(a) Player A's weight



(b) Weight ratio between Player A and an opponent

Second observation is a stochastic variation of weights assigned by the platform. This could be due to variation in the profile of user performing the query, hence their propensity of clicking into an ad. When a user located at Cambridge searched for the phrase "plumber", a local service provider would rightfully receive a much higher relevance score than one in America, for example. This variation in weights in effect adds noise to the advertiser's bid, so although her bids tend to remain constant or concentrate around a few values, the weighted bids are smoothed out. (see Figure 3.2)[1]

Therefore, this chapter is devoted to analysing the Bayes-Nash equilibria of the VCG and GSP, and also the dynamics of convergence of strategy profiles of different players.

[1] this provides additional grounds on our assumption in Chapter 1 and 2 that the distribution of bids admits a density.

3.2 Incomplete information setting for VCG and GSP

In VCG, the analysis of Bayes Nash equilibria follows very similarly to ex-post Nash. Truthful bidding is encouraged as before, and social utility is maximised. Therefore nothing much can be said about that.

For GSP, the mathematics is remarkably difficult, and very little is known. We aim to give a taste of the Bayes-Nash equilibria in very simple setting where analytic solutions are possible.

Consider the following: bids are submitted simultaneously and advertisers have private values per click which are position-independent. There are only 2 positions and 3 bidders. The click-through probabilities of the two slots are normalised to $(1, c_2)$, and the value per click of each bidder is i.i.d. drawn from the uniform distribution on the unit interval $v \sim U[0, 1]$.

Even this simple setting is sufficient to capture the dynamics and intuition of Bayes-Nash GSP, where the intensity of bid shading increases dramatically with c_2 (Figure 3.3):

Theorem 3.2.1. *[7, Gomes & Sweeney, p423, Proposition 1] Consider the GSP auction with $k = 2$ positions, $n = 3$ advertisers and click-through rate normalised to $(1, c_2)$, where the value per click of each bidder is an independent draw from the uniform distribution on the unit interval $v \sim U[0, 1]$. If a symmetric strategy exists, then it satisfies the following first-order linear differential equation subject to the boundary condition $\beta(0) = 0$:*

$$\beta'(v) = \frac{2v - 4vc_2 + c_2}{v(1 - c_2) + c_2} - \frac{1 - 2c_2}{v(1 - c_2) + c_2} \cdot \beta(v) \tag{3.1}$$

$\beta(v)$ is strictly increasing for all $v \in [0, 1]$ iff $c_2 \leq \frac{3}{4}$, so an efficient equilibrium exist only in this region.
In fact, the analytic solution is given by ($c_2 \neq \frac{1}{2}, \frac{2}{3}$):

$$\beta(v) = ((2 - c_2) \cdot v)^{\frac{1}{1-c_2}} \cdot \frac{B(c_2, v)}{A(c_2, v)} \tag{3.2}$$

where

$$A(c_2, v) := \frac{1}{6(c_2 - \frac{1}{2})(c_2 - \frac{2}{3})(\frac{(2-c_2)v}{c_2})^{\frac{1}{1-c_2}}}$$

$$B(c_2, v) := (c_2^{\frac{2c_2-1}{c_2^2-1}} + (-8c_2^{\frac{c_2}{c_2^2-1}} + 8c_2^{\frac{2c_2-1}{c_2^2-1}} + 2c_2^{\frac{1}{c_2^2-1}}) \cdot v - c_2^{\frac{4c_2-2}{c_2^2-1}} \cdot ((2 - c_2) \cdot v)^{\frac{2c_2-1}{1-c_2}}) \tag{3.3}$$

For $c_2 = \frac{1}{2}$,

$$\beta(v) = \log(1+v) \tag{3.4}$$

For $c_2 = \frac{2}{3}$,

$$\beta(v) = 3v + 4\log(2) + 2v\log(2) - 4log(2+v) - 2v\log(2+v) \tag{3.5}$$

We will only prove the first part of the theorem where $\beta(v)$ satisfies the *Equation 3.1*. The technique here can be easily generalised to any number of slots and bidders, and to different distributions of the valuation.

Proof. Let $z_l(v)$ denote the probability that a bidder with value v obtains slot l.

$$\begin{aligned} z_1(v) &= v^2 \\ z_2(v) &= 2(1-v)v \\ z_3(v) &= (1-v)^2 \end{aligned} \tag{3.6}$$

Let $P^E(v)$ be the payment of the bidder in equilibrium with value per click v. As it is a Bayes-Nash equilibrium, $\mathbb{E}(P^E(v))$ has to satisfy:

$$v \in \arg\max_{\hat{v}} \Big\{ \sum_{l=1}^{n} c_l z_l(\hat{v}) \cdot v - \mathbb{E}(P^E(\hat{v})) \Big\} \tag{3.7}$$

(v is chosen to maximise (expected click-through rate $\times$ value - expected payment))

We quote the *Envelope Theorem* (Theorem 3.2.4), which states that:

$$\sum_{l=1}^{n} c_l z_l(v) \cdot v - \mathbb{E}(P^E(v)) = -\mathbb{E}(P^E(0)) + \sum_{l=1}^{n} c_l \cdot \int_0^v z_l(t)dt \tag{3.8}$$

where $\mathbb{E}(P^E(0)) = 0$.

So the expected payment is given by:

$$\mathbb{E}(P^E(v)) = \sum_{l=1}^{n} c_l \cdot (z_l(v)v - \int_0^v z_l(t)dt) = \sum_{l=1}^{n} c_l \int_0^v \frac{dz_l(t)}{dt} t dt = \int_0^v 2t^2 dt + c_2 \int_0^v (2-4t)t dt \tag{3.9}$$

where the second equality follows from integration by parts.

Now suppose we are in an efficient equilibrium and all bidders play symmetric bidding strategy $\beta(v)$. Let $v^{j:k}$ be the j-th highest instance among $k \geq j$ iid

draws from our cdf $F(v) = v, v \in [0,1]$. From the rules of the GSP, we know that the expected payment of a bidder with private value v is given by:

$$\begin{aligned}
\mathbb{E}(P^{GSP}(v)) &= \sum_{l=1}^{n} c_l z_l(v) \mathbb{E}[\beta(v_{s+1:N}) | v^{s+1:N} \leq v \leq v^{s-1:N}] \\
&= z_1(v)\mathbb{E}[\beta(v^{2:3}) | v^{2:3} \leq v] + c_2 z_2(v) \mathbb{E}[\beta(v^{3:3} | v^{3:3} \leq v \leq v^{1:3}] \\
&= z_1(v)\mathbb{E}[\beta(v^{1:2} | v^{1:2} \leq v)] + c_2 z_2(v) \mathbb{E}[\beta(v^{1:1} | v^{1:1} \leq v] \\
&= \int_0^v \beta(t) 2t dt + 2(1-v)c_2 \int_0^v \beta(t) dt
\end{aligned}$$

Since $\mathbb{E}(P^E(v)) = \mathbb{E}(P^{GSP}(v))$ for all v,

$$\int_0^v 2t^2 dt + c_2 \int_0^v (2-4t) t dt = \int_0^v \beta(t) 2t dt + c_2 \int_0^v 2(1-v)\beta(t) dt \tag{3.10}$$

Differentiating with respect to v twice:

$$2v^2 + c_2(2-4v)v = \beta(v)2v + \beta(v)c_2 2(1-v) + \int_0^v -2c_2\beta(t)dt \tag{3.11}$$

$$4v + c_2(2-8v) = \beta'(v)(2v + 2c_2(1-v)) + \beta(v)(2-4c_2) \tag{3.12}$$

$$(v + c_2(1-v))\beta'(v) = (2v - 4vc_2 + 1) - (1-2c_2)\beta(v) \tag{3.13}$$

And we retrieve our differential equation. □

This has the bizarre result that an increase in click-through rate of lower slots in fact reduces bids from advertisers, by giving them higher incentive to shade their bid to go for a lower slot (see Figure 3.4). [2]

Interestingly, by setting an optimal reserve price this problem can be alleviated. This is captured by the following theorem which we are not going to prove.

Theorem 3.2.2. *[7, p430, Proposition 5] For the GSP auction with k positions and n bidders with private value i.i.d. with distribution F satisfying the increasing hazard rate condition, we can ensure that the platform's expected revenue increases with the click-through rate of all positions $l = 1, ..., k$ by setting the revenue-maximising reserve price r^* such that:*

$$r^* = \frac{1 - F(r^*)}{f(r^*)} \tag{3.14}$$

for all $l = 1, ..., n$.

[2] In the spirit of Bulow and Roberts (1989), we can draw analogy of this in terms of monopolistic price theory. He showed that a monopoly seller offering S products of heterogeneous qualities $c_1 \geq ... \geq c_s$, depending on the demand condition (elasticity of the demand curve), may suffer from reduced profit as the quality of some product $s \in \{2, ..., S\}$ increases.[7, p429]

Finally, what happens in most cases when the bidding function β is not monotonic, and an efficient equilibrium fails to exist? One might expect the existence of a *symmetric inefficient equilibrium*, where bidders use symmetric strategies, but the source of allocative inefficiency comes from when players randomise or pool on certain bids. In fact this is not possible, no symmetric strategy can exist at all, both mixed or pure strategies.

Theorem 3.2.3. *All symmetric equilibria of the GSP are outcome equivalent to a pure strategy equilibria.*

For all games satisfying the *single-crossing condition* as defined in *Milgrom and Shannon (1994)*, there exists a pure strategy equilibria. Proof is given in [13]. Intuitively it means that if a type v prefers a higher bid b' to a lower bid b, then so do all higher types $v' > v$, a condition satisfied by the GSP.

This leaves us with the possibility of *pooling*, i.e. for every type in an interval say $[\overline{w}, \underline{w}]$ where $\overline{w} \neq \underline{w}$, we submit the same bid $\hat{\beta}$. A proof [7, p432, Proposition 6] is beyond the scope of this paper so here is an intuition of why it cannot happen.
Suppose a bidder is bidding $\hat{\beta}$ in a pooling region $[\overline{w}, \underline{w}]$. If she is the only one who has a type within the region, then $\hat{\beta}$ yields the same payoff as either slightly overbidding $\hat{\beta} + \epsilon$ or underbidding $\hat{\beta} - \epsilon$. So the benefit of bidding $\hat{\beta}$ must be the opportunity to randomise among other players in a tie and hope to be chosen last so that she is competing with types below v, getting a lower position but paying less - for otherwise she can slightly overbid to secure the top place. However she could have guaranteed this competition by underbidding $\hat{\beta} - \epsilon$ at the start.

Theorem 3.2.4. *Envelope theorem. Suppose that $u(x, \cdot) : [0, 1] \to \mathbb{R}$ has the properties that:*

- *there exists a real-valued function $u_2(x, t)$ such that for all $x \in X$ and every $[a, b] \in [0, 1]$, $u(x, b) - u(x, a) = \int_a^b u_2(x, s)ds$, and*
- *there exists an integrable function $b : [0, 1] \to \mathbb{R}_+$ such that $|u_2(x, t)| \leq b(t)$ for all $x \in X$ and almost all $t \in [0, 1]$.*

Further suppose that $X^*(t) := \arg\max_{x \in X} u(x, t) \neq \emptyset$ for almost all $t \in [0, 1]$. Then for any selection $x^*(t) \in X^*(t)$,

$$V(t) = u(x^*(t), t) = u(x * (0), 0) + \int_0^t u_2(x^*(s), s)ds \tag{3.15}$$

Proof. First show that V is absolutely continuous, i.e. for every $\epsilon > 0$ there exists δ such that whenever a finite sequence of pairwise disjoint subintervals (x_k, y_k) has small total length: $\sum_k (y_k - x_k) < \delta$, then $\sum_k |V(y_k) - V(x_k)| < \epsilon$.

Let $B(t) = \int_0^t b(s)ds$.
For any $t', t'' \in [0,1]$ with $t' < t''$,

$$\begin{aligned}
|V(t'') - V(t')| &\leq \sup_{x \in X} |u(x,t'') - u(x,t')| \\
&= \sup_{x \in X} \Big| \int_{t'}^{t''} u_2(x,t)dt \Big| \\
&\leq \int_{t'}^{t''} \sup_{x \in X} |u_2(x,t)|dt \\
&\leq \int_{t'}^{t''} b(t)dt = B(t'') - B(t')
\end{aligned}$$

Fix $\epsilon > 0$. Select disjoint intervals (a_i, b_i). Let $\overline{x}_i \in X^*(b_i)$ and $\underline{x}_i \in X^*(a_i)$. $|V(b_i) - V(a_i)| = |u(\overline{x}_i, b_i) - u(\underline{x}_i, a_i)| \leq \max\{u(\overline{x}_i, b_i) - u(\overline{x}_i, a_i), u(\underline{x}_i, a_i) - u(\underline{x}_i, b_i)$.
This gives us

$$\begin{aligned}
\sum_{i=1}^k |V(b_i) - V(a_i)| &= \sum_{i=1}^k |u(\overline{x}_i, b_i) - u(\underline{x}_i, a_i)| \\
&\leq \sum_{i=1}^k \max(|u(\overline{x}_i, b_i) - u(\overline{x}_i, a_i)|, |u(\underline{x}_i, b_i) - u(\underline{x}_i, a_i)|) \\
&= \sum_{i=1}^k \max(|\int_{a_i}^{b_i} u_2(\overline{x}_i, t)dt|, |\int_{a_i}^{b_i} u_2(\underline{x}_i, t)dt|) \\
&\leq \sum_{i=1}^k \max(\int_{a_i}^{b_i} |u_2(\overline{x}_i, t)|dt, \int_{a_i}^{b_i} |u_2(\underline{x}_i, t)|dt) \\
&\leq \sum_{i=1}^k \int_{a_i}^{b_i} b(t)dt
\end{aligned}$$

b is integrable so we can find m such that $\int_{\{|b(t)|>M\}} |b(t)|dt < \epsilon/2$. Select $\delta < \epsilon/2M$ and we have

$$\sum_{i=1}^k \int_{a_i}^{b_i} b(t)dt < \epsilon/2 + M \sum_{i=1}^k |b_i - a_i| < \epsilon \tag{3.16}$$

Now V is absolutely continuous and hence differentiable almost everywhere. Supper t is a point of differentiability, $V(t) = u(x^*(t), t), V(t') \geq u(x^*(t), t')$, $V(t') - V(t) \geq u(x^*(t), t') - u(x^*(t), t)$. So

$$\frac{V(t') - v(t)}{|t' - t|} \geq \frac{u(x^*(t), t') - u(x^*(t), t)}{t' - t} \tag{3.17}$$

V is differentiable at t, letting $t' \downarrow t$ gives $V'(t) \geq u_2(x^*(t), t)$; and letting $t' \uparrow t$ gives $-V'(t) \geq u_2(x^*(t), t)$. Therefore $V'(t) = u_2(x^*(t), t)$. Finally apply the fundamental theorem of calculus to get the desired result. □

Figure 3.3: The bidding function for $c_2 \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ when three bidders with uniformly distributed values compete for two positions. Bid function is no longer monotonic for $c_2 \geq 3/4$ and we do not have an efficient equilibrium. Note the intense bid shading effect as c_2 approaches 1.

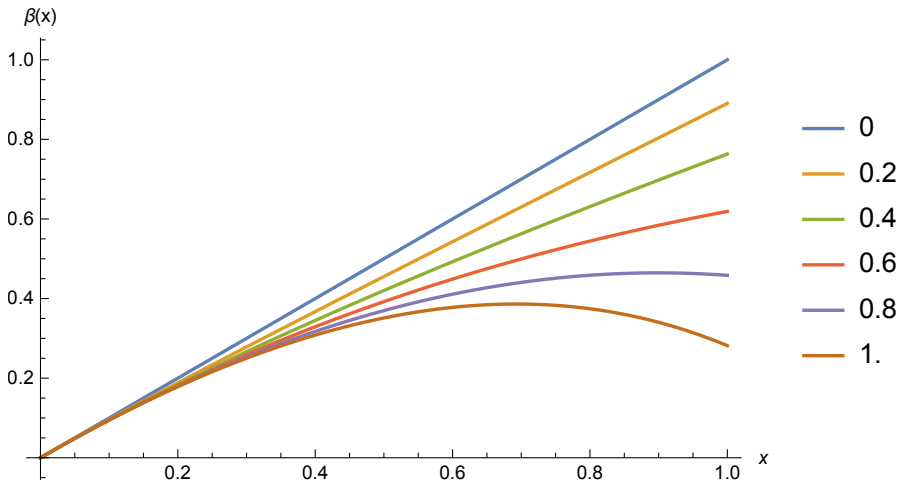
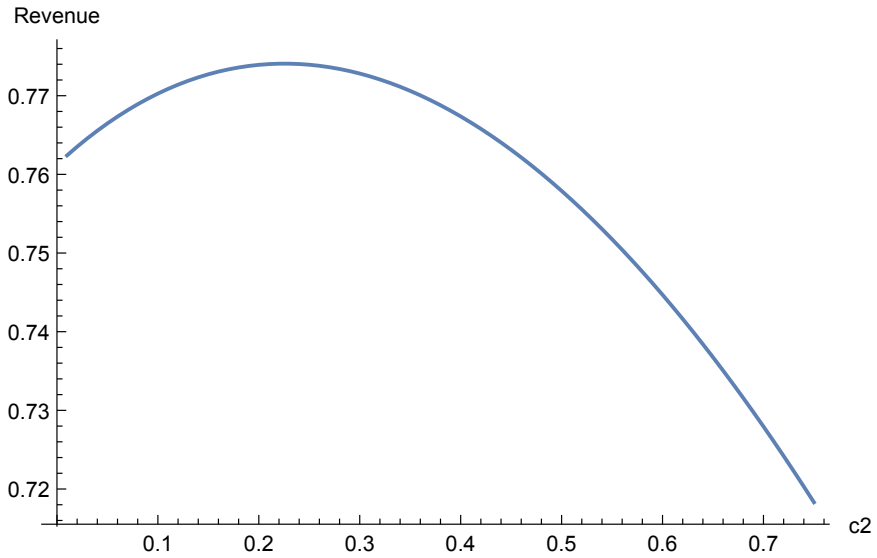


Figure 3.4: Seller revenue when three players bid optimally for two slots with click-through rate $(1, c_2)$ under GSP mechanism. Note that as the click-through rate of lower slot goes up, seller revenue decreases. Our analysis fails for $c_2 \geq 3/4$.



Chapter 4

Extension

Correlation between relevance and click-through probability:
In *Chapter 1* we decompose the click-through probability into "position effect" α_l and "relevance effect" p_i^τ. In reality they not be independent. VCG model has the added advantage of handling generalised situations like this very eloquently. Paper from Kelly et al. [1, Section 6 & 7] gave a framework for analysing VCG model and its implementation. His method may be generalised to advertisers having different preferences for slot positions (so even more general than the user heterogeneity we looked at). But importantly we require the platform to have full knowledge of the "real" click-through probabilities.

Reserve price:
When the platform decides on which mechanism they use, they look at the allocative efficiency and also importantly the revenue generated. Optimal auction theory tells us that seller's revenue is maximised by putting a reserve price for the slots (or equivalently a participation fee for each bidder) [12, Lecture 22]. This necessarily changes bidder participation and hence jeopardises allocative efficiency, for the slots are not utilised if none of the bids meet the reserve price. It also has the property that the winner is not necessarily the highest bidder.
In the incomplete information setting where each player has their own distribution of private value, the reserve price is worked out by the unique solution of

$$r^* = \frac{1 - F(r^*)}{r^*} \tag{4.1}$$

This assumes an increasing hazard rate, which is satisfied by most common distributions. In practice, most platforms impose reserve prices[1].
Robert et al. [3] investigated the effect of using reserve price not only as a threshold to filter out lower bids, but also incorporating it into the ranking

[1] Google: `https://www.en.advertisercommunity.com/t5/Basics-for-New-Advertisers/Minimum-Bid/td-p/428552`

function. His focus is more on maximising revenue than social welfare.

Budget-constraint optimisation:
Recently Facebook started offering bidding bot services for advertisers with a fixed budget constraint aimed to maximise the number of clicks received[2]. This has proven to be popular as very often advertisers are unaware of the marginal utility of an extra click, but rather assign a fixed budget for online advertising.

There is no simple way of converting this optimisation problem directly to our previous analysis with the use of a utility function. But in general there are two different ways of looking at it.

The method adopted by Tan & Srikant [11] is to assume that advertisers just pay a known constant price per click, and do not adjust to the bids. There is the additional constraint that they they operate within a limited budget. Tan & Srikant focused on how the platform should allocate adverts to search queries which arrive stochastically. The platform wishes to maximise the average revenue to as close to the budget constraint as possible, without going over. Turns out the process for each advertiser may be modelled as a queueing process with constant serving rate (the average budget) and exponential arrival rate (assignment of adverts which lead to a click).

Tran-Thanh et al. [10] focused on the incomplete information setting as described in *Chapter 2*. When an advertiser enters a new market, she has no information on the market price distribution. The only way to gain such information is by bidding[3]. She has to balance between bidding for information on market price (*exploration*), and using this knowledge to maximise the number of clicks she gets (*exploitation*), all being done while operating under a limited budget. Recently a lot of development has been made by machine learning, adopting algorithms from other fields of operational research. Tran-Thanh et al. investigated the use of the online knapsack problem. In this problem, an optimiser is presented with a sequence of items with values and weights, and at each time step he has to make an irrevocable decision of taking the item or giving it up. This has analogy in advert assignment problem in the sense that at each auction an advertiser has the chance of taking a slot (thereby reducing the remaining future budget) or giving it up (saving money for future opportunities). These sort of algorithms yield good result when applied to real life distribution and is a rapidly growing area of study.

[2] `https://www.facebook.com/business/help/527780867299597/`

[3] this is sometimes known as a *right-censored auction*. Winning player knows their bid and the second highest bid (it would be the price they pay). However, losing players are not told the winning bid, they only have the knowledge that the winning bid is above theirs.

Dynamics of convergence:
Between the incomplete information setting depicted in *Chapter 3* and the complete information setting in *Chapter 1* lies the issue of convergence dynamics. Kelly et al. [1, p10] looked at advertisers who has delayed feedback on the click-through rate she has achieved with her bid. They showed that under the VCG mechanism and certain monotonicity condition, convergence to the Nash equilibrium must occur. In the GSP setting not a lot of research has been done, and it would be an interesting direction for future research.

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